The New Keynesian Model

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Advanced Macroeconomics – Part 2 – Lecture 2 and 3

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Roadmap

- Introduction: from the Old Keynesian (IS–LM, AS–AD) to the NK model
- Building the New Keynesian model
  - households, monopolistic competition, Calvo pricing
  - non-linear model first, then log-linearization
- Determinacy and Taylor principle
- Monopolistic competition and aggregate demand externality (Blanchard–Kiyotaki)
- Transmission of shocks in the 3-equation model
- Empirics of the building blocks: Euler equation, price rigidity, Phillips curve
- Evidence on monetary non-neutrality
- Policy: divine coincidence, optimal policy, monetary vs. fiscal policy, forward guidance

Introduction

Why a New Keynesian model?

- In a nutshell: NK model = RBC model + monopolistic competition + nominal rigidities
- RBC / neoclassical model: classical dichotomy
  - real allocation determined independently of nominal variables ⇒ money is neutral
  - fluctuations are efficient ⇒ no role for stabilization policy
- Yet, in the data:
  - prices and wages are sticky: median duration of prices 8–11 months (Nakamura & Steinsson, 2008), wages ≈ 1 year (Taylor, 1999)
  - monetary policy has real effects (Christiano, Eichenbaum & Evans, 1999; Romer & Romer, 2004; Nakamura & Steinsson, 2018)
- Sticky prices break monetary neutrality *and* the first welfare theorem
  ⇒ rationale for stabilization policy
- ***"New"*** = methodological: microfounded, rational expectations
  "Keynesian" = substantive: aggregate demand matters in the short run
- Workhorse at central banks: Fed (FRB/US, EDO), ECB (NAWM), Smets-Wouters (2007)

The Old Keynesian model: IS–LM

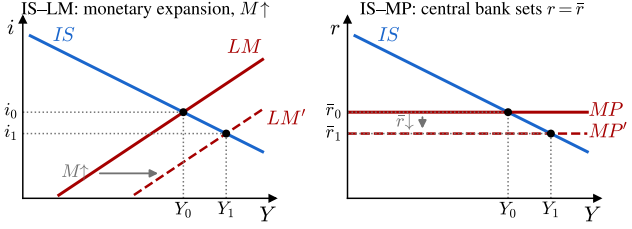
- Hicks (1937) formalization of Keynes (1936). Prices are fixed in the short run
- ***IS*** (goods market):

$$Y = C(Y - T) + I(r) + G, \qquad 0 < C' < 1, \quad I' < 0, \qquad r = i - \pi^e$$

- ***LM*** (money market):

$$\frac{M}{P} = L(i, Y), \qquad L_i < 0, \quad L_Y > 0$$

- Output is demand-determined: Keynesian cross, multiplier $\frac{1}{1-C'} > 1$
- Modern variant (Romer, 2000): replace LM by central bank interest rate rule (MP curve)

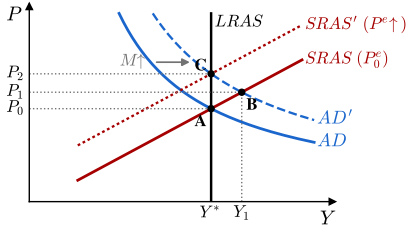


The Old Keynesian model: AS–AD and the Phillips curve

- ***AD***: combine IS and LM, eliminate $i \Rightarrow\ Y = Y^D(M/P, G, T)$, decreasing in P
- ***AS***: expectations-augmented Phillips curve (Friedman, 1968; Phelps, 1968)

$$\pi_t = \pi_t^e + \kappa\left(Y_t - Y_t^*\right) + u_t$$

  with adaptive expectations $\pi_t^e = \pi_{t-1}$: accelerationist Phillips curve
- Short run: upward-sloping AS; long run: vertical at potential Y^*
- Problems:
  - behavioral equations not derived from optimization $\Rightarrow$ not policy-invariant (Lucas, 1976)
  - backward-looking expectations $\Rightarrow$ systematically exploitable trade-off; 1970s' stagflation
  - no welfare criterion



Target of today: the three-equation NK model

$$
\begin{aligned}
&\text{(DIS)} && x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma}\left(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n\right) \\
&\text{(NKPC)} && \pi_t = \beta\, \mathbb{E}_t \pi_{t+1} + \kappa\, x_t \\
&\text{(MP)} && i_t = \rho + \phi_\pi\, \pi_t + \phi_y\, x_t + \nu_t
\end{aligned}
$$

- $x_t = y_t - y_t^n$: output gap w.r.t. the *natural* (flexible-price) level of output
- Modern version of IS–LM, AS–AD:

Old Keynesian	**New Keynesian**	**Microfoundation**
IS curve	Dynamic IS (DIS)	household Euler equation
LM curve	interest rate rule (MP)	central bank (Taylor) rule
AS / Phillips curve	NK Phillips curve (NKPC)	Calvo price setting

- DIS & NKPC are forward-looking: what matters is the *entire path* of expected policy

Building the New Keynesian Model

Ingredients

1. ***Representative household*** (key! – relaxed in HANK lectures)
   - consumes a CES basket of varieties, supplies labor, trades a nominal bond
2. ***Production*** with nominal price rigidities
   - continuum of monopolistically competitive firms (Dixit–Stiglitz)
   - Calvo (1983) pricing: random fraction $1-\theta$ of firms can reset their price
   - alternative: Rotemberg (1982) quadratic adjustment costs: same Phillips curve to first order
3. ***Monetary policy***: central bank sets the nominal rate i_t through a rule (Taylor rule)

- Simplifications: no capital, no explicit money: cashless limit (Woodford, 2003)
  $\Rightarrow$ a "neo-Wicksellian" model: the price *level* is a nominal anchor problem
- Strategy: write non-linear model, then log-linearize around zero-inflation steady state

Households: demand for varieties

- Continuum of varieties $i \in [0,1]$, Dixit–Stiglitz aggregator with elasticity $\varepsilon > 1$:

$$C_t = \Big(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \, di \Big)^{\frac{\varepsilon}{\varepsilon-1}} \qquad s.t. \int_0^1 P_t(i) C_t(i) \, di = I_t$$

- Lagrangian $\mathcal{L}\big(\{C_t(i)\}, \lambda\big) = \Big(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} \, di \Big)^{\frac{\varepsilon}{\varepsilon-1}} - \lambda \Big(\int_0^1 P_t(i) C_t(i) \, di - I_t \Big)$
- Solution: isoelastic demand curves and price index

$$C_t(i) = \Big(\frac{P_t(i)}{P_t} \Big)^{-\varepsilon} C_t, \qquad P_t = \Big(\int_0^1 P_t(i)^{1-\varepsilon} \, di \Big)^{\frac{1}{1-\varepsilon}}, \qquad \int_0^1 P_t(i) C_t(i) \, di = P_t C_t = I_t$$

- ε = elasticity of substitution btw varieties = price elasticity of demand faced by each firm

Households: intertemporal problem

- Representative household:

$$\max_{\{C_t,N_t,B_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \Big]$$

$$\text{s.t.} \qquad P_t C_t + B_t \leq (1+i_{t-1}) B_{t-1} + W_t N_t + D_t - T_t$$

  B_t: one-period nominal bond, D_t: dividends from firms, T_t: lump-sum taxes
  σ: inverse EIS; φ: inverse Frisch elasticity

- Two first-order conditions (+ no-Ponzi / transversality):

$$\text{Euler:} \qquad C_t^{-\sigma} = \beta \, \mathbb{E}_t \Big[(1+i_t) \frac{P_t}{P_{t+1}} C_{t+1}^{-\sigma} \Big]$$

$$\text{Labor supply:} \qquad \frac{W_t}{P_t} = N_t^{\varphi} \, C_t^{\sigma}$$

- Inflation $\Pi_t \equiv P_t / P_{t-1}$; nominal SDF $\Lambda_{t,t+k} = \beta^k \big(\frac{C_{t+k}}{C_t} \big)^{-\sigma} \frac{P_t}{P_{t+k}}$

Firms: technology and the flexible-price benchmark

- Firm i: $Y_t(i) = Z_t N_t(i)$, faces demand $Y_t(i) = (P_t(i)/P_t)^{-\varepsilon} Y_t$
- Nominal marginal cost W_t/Z_t, common to all firms; real marginal cost $mc_t \equiv \frac{W_t}{P_t Z_t}$
- ***Flexible prices***: $\max_{P(i)} \left(P(i) - W_t/Z_t\right)\left(P(i)/P_t\right)^{-\varepsilon} Y_t$

$$\Rightarrow \quad P_t(i) = \mu \frac{W_t}{Z_t}, \qquad \mu \equiv \frac{\varepsilon}{\varepsilon - 1} > 1 \qquad \Rightarrow \qquad mc_t = \frac{1}{\mu}$$

- Equilibrium real block: labor supply $W_t/P_t = N_t^{\varphi} C_t^{\sigma}$; $mc_t = 1/\mu$; $Y_t = Z_t N_t$; $C_t = Y_t$

$$\Rightarrow \qquad \left(Y_t^n\right)^{\sigma+\varphi} = \frac{Z_t^{1+\varphi}}{\mu} \qquad \text{(natural output)}$$

  and Euler equation gives natural real rate r_t^n

- Real allocation independent of i_t, M_t, P_t: classical dichotomy
  - monetary policy only determines nominal variables (and with a peg, not even P_t)
  - still inefficient: markup $\mu > 1 \Rightarrow Y_t^n$ below the efficient level

Why sticky prices matter: a two-period illustration

- Two periods, $Y_1 = A_1K_1$, $Y_2 = A_2K_2$, $K_2 = I_1$, money quantity $M_t = P_tC_t$, policy (i_1, M_2)
  Here $\eta = 1/\sigma$ = EIS: Euler equation: $C_1 = C_2[\beta(1+r_1)]^{-\eta}$
- ***Flexible*** P_1: $Y_1 = A_1K_1$, $1 + r_1 = A_2$: monetary policy only moves P_1, P_2
- ***Sticky*** P_1 (set in advance): output determined by the short side, $Y_1 = \min\{Y_1^D, Y_1^S\}$

$$Y_1 = \left[1 + \left(\frac{1}{\beta A_2}\right)^{\eta} A_2\right] \frac{M_2}{(1+i_1)P_1}$$

  - $i_1 \downarrow \Rightarrow C_1, I_1, Y_1 \uparrow$: monetary non-neutrality
  - demand shocks (news $A_2 \downarrow$, patience $\beta \uparrow$) $\Rightarrow$ recession (with $\eta < 1$)
- Key mechanism: household income depends on aggregate demand, $\Omega = P_1Y_1 = P_1Y_1^D$
  "one person's spending is someone else's income" $\Rightarrow$ Keynesian-cross feedback
- Paradox of thrift: $\beta \uparrow \Rightarrow C_1 \downarrow, Y_1 \downarrow$ but I_1 unchanged: saving does not rise

Calvo pricing: the reset price problem

- Each period, a firm can reset its price with probability $1-\theta$, independently of history
  $\Rightarrow$ expected price duration $1/(1-\theta)$
- A resetting firm chooses P_t^* to maximize the value of profits while the price is in place:

$$\max_{P_t^*} \sum_{k=0}^{\infty} \theta^k \, \mathbb{E}_t \left[\Lambda_{t,t+k} \left(P_t^* \, Y_{t+k|t} - \frac{W_{t+k}}{Z_{t+k}} Y_{t+k|t} \right) \right]$$

$$\text{s.t.} \qquad Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\varepsilon} Y_{t+k}$$

- First-order condition:

$$\sum_{k=0}^{\infty} \theta^k \, \mathbb{E}_t \left[\Lambda_{t,t+k} \, Y_{t+k|t} \left(P_t^* - \mu \, \frac{W_{t+k}}{Z_{t+k}} \right) \right] = 0$$

- Reset price = markup μ over a weighted average of current and expected future nominal marginal costs
- $\theta = 0$: back to the static markup $P_t^* = \mu W_t / Z_t$

The non-linear Phillips curve

- Rearranging the FOC, with $\Pi_{t,t+k} \equiv P_{t+k}/P_t$:

$$p_t^* \equiv \frac{P_t^*}{P_t} = \frac{\mu\, \mathbb{E}_t \sum_{k\geq 0} (\beta\theta)^k C_{t+k}^{-\sigma} Y_{t+k}\, \Pi_{t,t+k}^{\varepsilon}\, mc_{t+k}}{\mathbb{E}_t \sum_{k\geq 0} (\beta\theta)^k C_{t+k}^{-\sigma} Y_{t+k}\, \Pi_{t,t+k}^{\varepsilon-1}} = \frac{F_t}{H_t}$$

- Recursively:

$$F_t = \mu\, C_t^{-\sigma} Y_t\, mc_t + \beta\theta\, \mathbb{E}_t\left[\Pi_{t+1}^{\varepsilon} F_{t+1}\right]$$
$$H_t = C_t^{-\sigma} Y_t + \beta\theta\, \mathbb{E}_t\left[\Pi_{t+1}^{\varepsilon-1} H_{t+1}\right]$$

- Price index: resetters are drawn at random and all choose P_t^*

$$P_t^{1-\varepsilon} = \theta P_{t-1}^{1-\varepsilon} + (1-\theta)(P_t^*)^{1-\varepsilon} \qquad \Rightarrow \qquad \left(\frac{1-\theta\,\Pi_t^{\varepsilon-1}}{1-\theta}\right)^{\frac{1}{1-\varepsilon}} = \frac{F_t}{H_t}$$

- Highest inflation possible ($P_t^* \to \infty$): $\Pi_t < \theta^{-1/(\varepsilon-1)}$
  standard calibration $\theta = 0.85$, $\varepsilon = 6$: $\Pi < 1.03$ per quarter!

Price dispersion and aggregate production

- Aggregate labor demand:

$$N_t = \int_0^1 N_t(i)\,di = \int_0^1 \frac{Y_t(i)}{Z_t}\,di = \frac{Y_t}{Z_t}\underbrace{\int_0^1 \left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon} di}_{\equiv \Delta_t} \qquad \Rightarrow \qquad Y_t = \frac{Z_t N_t}{\Delta_t}$$

- $\Delta_t \geq 1$ (Jensen): price dispersion $\Rightarrow$ misallocation of labor across varieties $\Rightarrow$ TFP loss (Straub writes $Y_t = \Delta_t Z_t N_t$ with $\Delta_t \leq 1$: same object, inverted)
- Law of motion:

$$\Delta_t = \theta\,\Pi_t^{\varepsilon}\,\Delta_{t-1} + (1-\theta)\left(\frac{1-\theta\,\Pi_t^{\varepsilon-1}}{1-\theta}\right)^{\frac{\varepsilon}{\varepsilon-1}}$$

- Around $\Pi = 1$: $\Delta_t - 1 = \mathcal{O}(\|\pi\|^2)$
  - second order: disappears in the log-linear model
  - but key for welfare: this is *the* cost of inflation in the NK model

Taking stock: the non-linear NK model

1. Household optimality:

$$C_t^{-\sigma} = \beta\,\mathbb{E}_t\Big[\frac{1+i_t}{\Pi_{t+1}}C_{t+1}^{-\sigma}\Big], \qquad w_t \equiv \frac{W_t}{P_t} = N_t^{\varphi}C_t^{\sigma}, \qquad mc_t = \frac{w_t}{Z_t}$$

2. Non-linear Phillips curve: $\big(\frac{1-\theta\Pi_t^{\varepsilon-1}}{1-\theta}\big)^{\frac{1}{1-\varepsilon}} = F_t/H_t$, with (F_t, H_t) as above
3. Output and price dispersion: $C_t = Y_t = Z_t N_t/\Delta_t$, Δ_t law of motion
4. Monetary policy rule, e.g. $1 + i_t = \frac{1}{\beta}\,\Pi_t^{\phi_\pi}\big(Y_t/Y_t^n\big)^{\phi_y} e^{v_t}$

- ***Steady state*** with zero inflation $\Pi = 1$:

$$p^* = 1, \qquad \Delta = 1, \qquad mc = \frac{1}{\mu}, \qquad 1 + i = \frac{1}{\beta}, \qquad Y = Y^n$$

- Can be solved non-linearly (Dynare, perfect foresight / sequence-space methods)
  $\Rightarrow$ useful for large shocks, high inflation, ZLB

Log-linearization I: households and marginal cost

- Lowercase = log-deviations from steady state; i_t, π_t = net rates; $\rho \equiv -\log \beta$
- Euler equation:
$$c_t = \mathbb{E}_t c_{t+1} - \frac{1}{\sigma}\left(i_t - \mathbb{E}_t \pi_{t+1} - \rho\right)$$
- Labor supply: $w_t = \sigma c_t + \varphi n_t$
- Production and goods market: $y_t = c_t$, $y_t = z_t + n_t$ (Δ_t drops out at first order)
- Real marginal cost:
$$\widehat{mc}_t = w_t - z_t = \sigma y_t + \varphi(y_t - z_t) - z_t = (\sigma + \varphi)\, y_t - (1 + \varphi)\, z_t$$
- Firms' marginal cost rises with output: higher labor $\Rightarrow$ higher wage (labor supply) and lower marginal utility of consumption (wealth effect)

Log-linearization II: the New Keynesian Phillips curve

- Log-linearize the reset-price FOC around the zero-inflation steady state:

$$p_t^* = (1-\beta\theta)\sum_{k=0}^{\infty}(\beta\theta)^k\,\mathbb{E}_t\left[\widehat{mc}_{t+k} + p_{t+k}\right]$$

  i.e. recursively: $p_t^* - p_{t-1} = \beta\theta\,\mathbb{E}_t\left[p_{t+1}^* - p_t\right] + (1-\beta\theta)\widehat{mc}_t + \pi_t$

- Price index: $\pi_t = (1-\theta)(p_t^* - p_{t-1})$
- Combining:

$$\boxed{\pi_t = \beta\,\mathbb{E}_t\pi_{t+1} + \lambda\,\widehat{mc}_t}, \qquad \lambda \equiv \frac{(1-\theta)(1-\beta\theta)}{\theta}$$

- Solving forward: inflation = present value of future real marginal costs

$$\pi_t = \lambda\sum_{k=0}^{\infty}\beta^k\,\mathbb{E}_t\,\widehat{mc}_{t+k}$$

- Purely forward-looking: firms raise prices today if they expect high future costs

Log-linearization III: output gap and natural rate

- Natural output (flexible prices $\Leftrightarrow \widehat{mc}_t = 0$):

$$y_t^n = \frac{1+\varphi}{\sigma+\varphi} z_t \qquad \Rightarrow \qquad \widehat{mc}_t = (\sigma+\varphi)\left(y_t - y_t^n\right) \equiv (\sigma+\varphi)\, x_t$$

- NKPC in terms of the output gap:

$$\pi_t = \beta\, \mathbb{E}_t \pi_{t+1} + \kappa\, x_t, \qquad \kappa \equiv \lambda(\sigma+\varphi)$$

- Natural real rate (Euler equation in the flex-price economy):

$$r_t^n = \rho + \sigma\, \mathbb{E}_t \Delta y_{t+1}^n = \rho + \sigma \frac{1+\varphi}{\sigma+\varphi}\, \mathbb{E}_t \Delta z_{t+1}$$

(+ demand / preference shocks shift r_t^n)

- Subtracting the two Euler equations: the Dynamic IS curve

$$x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma}\left(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n\right)$$

The three-equation NK model

$$\begin{aligned} x_t &= \mathbb{E}_t x_{t+1} - \sigma^{-1}\left(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n\right) && \text{(DIS)} \\ \pi_t &= \beta\, \mathbb{E}_t \pi_{t+1} + \kappa\, x_t \left[+ \; u_t \right] && \text{(NKPC)} \\ i_t &= \rho + \phi_\pi\, \pi_t + \phi_y\, x_t + v_t && \text{(MP)} \end{aligned}$$

- Two jump variables (x_t, π_t), no endogenous state (no backward-looking variable!)
- Shocks: r_t^n (technology, demand), v_t (monetary), u_t (cost-push / markup)
- Solving the DIS forward:

$$x_t = -\frac{1}{\sigma} \sum_{k=0}^{\infty} \mathbb{E}_t\left(r_{t+k} - r_{t+k}^n\right) + \lim_{T\to\infty} \mathbb{E}_t x_{t+T}, \qquad r_t \equiv i_t - \mathbb{E}_t \pi_{t+1}$$

  $\Rightarrow$ the output gap depends on the whole path of future real-rate gaps
- Budget constraint? Lump-sum taxes adjust: Ricardian equivalence

Old vs. New Keynesian: what changes?

- Output gap measured w.r.t. the natural level of output, which fluctuates
  $\Rightarrow$ not all fluctuations are bad (e.g. efficient response to TFP)
- Aggregate demand and supply depend on expectations of future interest rates and inflation
  $\Rightarrow$ central banking = managing expectations: communication, inflation targets, credibility

- Microfounded: deep parameters $(\beta, \sigma, \varphi, \varepsilon, \theta)$, welfare analysis
  (Lucas critique: is the Calvo θ structural? $\rightarrow$ menu costs later)
- But: the NKPC is purely forward-looking
  - no intrinsic inflation persistence; credible disinflations are costless (Ball, 1994)
  - fix: indexation of non-resetting prices to past inflation (Christiano, Eichenbaum & Evans, 2005):

$$\pi_t = \frac{\beta}{1+\beta}\mathbb{E}_t\pi_{t+1} + \frac{1}{1+\beta}\pi_{t-1} + \frac{\kappa}{1+\beta}x_t$$

  - Galí & Gertler (1999): forward-looking behavior dominant, backward-looking component significant but small